

```
#numpy-numpy is an open source library which is used for scientific calculations
```

```
import numpy as np
```

```
a=[1,2,3,4]  
type(a)
```

```
list
```

```
b=np.array(a)  
b
```

```
array([1, 2, 3, 4])
```

```
type(b)
```

```
numpy.ndarray
```

```
b.ndim
```

```
1
```

```
a=[[1,2,3],[4,5,6],[7,8,9]]  
a
```

```
[[1, 2, 3], [4, 5, 6], [7, 8, 9]]
```

```
b=np.array(a)  
b
```

```
array([[1, 2, 3],  
       [4, 5, 6],  
       [7, 8, 9]])
```

```
#matrix total element=n(row)*n(col)  
#row1=[1,2,3]  
#row2=[4,5,6]  
#row3=[7,8,9]  
  
#col1=[1,4,7]  
#col2=[2,5,8]  
#col3=[3,6,9]  
  
#total element=3*3=9
```

```
print('total row and col',b.shape)  
print('total rows',b.shape[0])  
print('total col',b.shape[1])  
print('total element in array',b.size)  
print('total dimension',b.ndim)
```

```
total row and col (3, 3)  
total rows 3  
total col 3  
total element in array 9  
total dimension 2
```

```
# difference b/w list and array  
a=[1,2,4]  
a*3
```

```
[1, 2, 4, 1, 2, 4, 1, 2, 4]
```

```
a=np.array([1,2,4])  
print(a)  
print(a*3)
```

```
[1 2 4]  
[ 3  6 12]
```

```
#pre define fun in numpy
#image=pixels=0-255=normalize=0-1
#if i want to add black shadow then i have to use 0
#if i want to add white shadow then i have to use 1
```

#1.zeros()-it will create an array in which all the element will be 0 in either one dimension or multi-dim.

```
a=np.zeros(3)
a
array([0., 0., 0.])
```

```
a=np.zeros((3,4))
a
array([[0., 0., 0., 0.],
       [0., 0., 0., 0.],
       [0., 0., 0., 0.]])
```

#2.ones()-it will create an array in which all the element will be 1 in either one dimension or multi-dim.

```
a=np.ones(3)
a
array([1., 1., 1.])
```

```
a=np.ones((2,3))
a
array([[1., 1., 1.],
       [1., 1., 1.]])
```

#3.eye()-it will create an array in which diagonal position element will be 1 and rest all 0

```
a=np.eye(4)
a
array([[1., 0., 0., 0.],
       [0., 1., 0., 0.],
       [0., 0., 1., 0.],
       [0., 0., 0., 1.]])
```

#4.diag()-it will create an array in which we can set our custom value on diagonal position

```
a=np.array([1,45,78,90])
a
array([ 1, 45, 78, 90])
```

```
b=np.diag(a)
b
array([[ 1,  0,  0,  0],
       [ 0, 45,  0,  0],
       [ 0,  0, 78,  0],
       [ 0,  0,  0, 90]])
```

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